

Class-12: Electron MCQS

Chapter: Electrons — > 100 MCQ Questions

Useful constants:

$$e = 1.6 \times 10^{-19} C$$

$$m_e = 9.1 \times 10^{-31} kg$$

$$\frac{e}{m_e} = 1.76 \times 10^{11} C kg^{-1}$$

$$g = 9.8 m/s^2$$

Quickly recall:

$$F = qE$$

$$F = qvB \sin \theta$$

$$v = \frac{E}{B}$$

$$\frac{e}{m} = \frac{v^2}{Br}$$

$$K = eV$$

$$v = \sqrt{\frac{2eV}{m}}$$

$$r = \frac{mv}{eB}$$

$$\omega = \frac{eB}{m}$$

$$T = \frac{2\pi m}{eB}$$

$$f = \frac{eB}{2\pi m}$$

$$q = ne$$

Section A: Discovery and Fundamental Properties

1.

The electron was discovered by:

- A. Rutherford
- B. J. J. Thomson
- C. Millikan
- D. Chadwick

2.

The charge-to-mass ratio (e/m) of an electron was determined by:

- A. Millikan
- B. J. J. Thomson
- C. Rutherford
- D. Bohr

3.

The charge of an electron was measured by:

- A. J. J. Thomson
- B. R. A. Millikan
- C. Rutherford
- D. Goldstein

4.

The magnitude of charge of an electron is:

- A. $1.6 \times 10^{-19} C$
- B. $9.1 \times 10^{-31} C$
- C. $1.76 \times 10^{11} C$
- D. $6.63 \times 10^{-34} C$

5.

The mass of an electron is approximately:

- A. $1.67 \times 10^{-27} kg$
- B. $9.1 \times 10^{-31} kg$
- C. $1.6 \times 10^{-19} kg$
- D. $9.1 \times 10^{-19} kg$

6.

The specific charge of an electron is approximately:

- A. $1.76 \times 10^{11} C/kg$
- B. $9.1 \times 10^{-31} C/kg$
- C. $1.6 \times 10^{-19} C/kg$
- D. $1.76 \times 10^{-19} C/kg$

7.

The charge on an electron is:

- A. Positive
- B. Negative

- C. Neutral
- D. Sometimes positive

8.

The charge on a proton compared with that on an electron is:

- A. Equal in magnitude and same in sign
- B. Equal in magnitude and opposite in sign
- C. Greater in magnitude
- D. Smaller in magnitude

9.

The approximate ratio of proton mass to electron mass is:

- A. 1837
- B. $1/1837$
- C. 938
- D. 1

10.

The smallest known unit of free electric charge is:

- A. Proton charge
- B. Electron charge
- C. Neutron charge
- D. Atomic charge

Section B: Cathode Rays and Discharge Tube

11.

Cathode rays originate from the:

- A. Anode
- B. Cathode
- C. Gas molecules
- D. Glass wall

12.

Cathode rays are composed of:

- A. Protons
- B. Neutrons

- C. Electrons
- D. Positive ions

13.

Cathode rays travel:

- A. From anode to cathode
- B. From cathode to anode
- C. In both directions
- D. Randomly

14.

Cathode rays are deflected by:

- A. Electric field only
- B. Magnetic field only
- C. Both electric and magnetic fields
- D. Neither field

15.

Cathode rays are deflected toward the positive plate in an electric field because they are:

- A. Positively charged
- B. Negatively charged
- C. Neutral
- D. Uncharged particles

16.

The nature of cathode rays is:

- A. Dependent on the gas used
- B. Dependent on the electrode
- C. Independent of the gas and electrode
- D. Dependent only on pressure

17.

Cathode rays produce fluorescence when they strike:

- A. The cathode
- B. The anode
- C. The fluorescent glass wall
- D. The gas molecules only

18.

Cathode rays travel in:

- A. Curved paths naturally
- B. Straight lines in the absence of fields
- C. Circular paths
- D. Elliptical paths

19.

The pressure inside a discharge tube is reduced mainly to:

- A. Increase collisions
- B. Reduce collisions
- C. Stop electron emission
- D. Increase atmospheric pressure

20.

The high potential difference in a discharge tube is required mainly to:

- A. Heat the glass
- B. Accelerate charged particles
- C. Produce light only
- D. Increase atmospheric pressure

Section C: Thomson's e/m Experiment

21.

Thomson determined the:

- A. Charge of electron
- B. Mass of electron directly
- C. Specific charge of electron
- D. Charge of proton

22.

In Thomson's experiment, the electric and magnetic fields are:

- A. Parallel
- B. Perpendicular
- C. Antiparallel
- D. At 45°

23.

When an electron beam passes undeflected through crossed electric and magnetic fields:

- A. $eE = evB$
- B. $E = vB$
- C. $v = E/B$
- D. All of the above

24.

The velocity of an electron beam passing undeflected through crossed fields is:

- A. EB
- B. E/B
- C. B/E
- D. $E + B$

25.

If $E = 2 \times 10^4 \text{ N/C}$ and $B = 4 \times 10^{-3} \text{ T}$, the velocity of the undeflected electron beam is:

- A. $5 \times 10^6 \text{ m/s}$
- B. $8 \times 10^6 \text{ m/s}$
- C. $5 \times 10^7 \text{ m/s}$
- D. $8 \times 10^7 \text{ m/s}$

26.

In Thomson's experiment, if the electric field is removed, the electron beam is deflected by:

- A. Electric force
- B. Magnetic force
- C. Gravitational force only
- D. No force

27.

The magnetic force on a moving electron is maximum when its velocity is:

- A. Parallel to B
- B. Antiparallel to B
- C. Perpendicular to B
- D. Zero

Answer: C

28.

The magnetic force on an electron moving parallel to a magnetic field is:

- A. Maximum
- B. Minimum but non-zero
- C. Zero
- D. Infinite

29.

The relation used to determine e/m in Thomson's experiment after measuring velocity and radius is:

- A. $e/m = vBr^{-1}$
- B. $e/m = v/(Br)$
- C. $e/m = v^2/(Br)$
- D. $e/m = Br/v^2$

30.

The value of e/m for an electron is approximately:

- A. $1.76 \times 10^{11} C/kg$
- B. $1.6 \times 10^{-19} C/kg$
- C. $9.1 \times 10^{-31} C/kg$
- D. $6.67 \times 10^{-11} C/kg$

Section D: Electron in Electric Field

31.

The force on an electron in an electric field E is:

- A. e/E
- B. eE
- C. E/e
- D. eE^2

32.

The direction of force on an electron in an electric field is:

- A. Along the field
- B. Opposite to the field
- C. Perpendicular to the field
- D. Random

33.

The acceleration of an electron in a uniform electric field is:

- A. eE/m
- B. mE/e
- C. $e/(mE)$
- D. m/e

34.

An electron projected horizontally into a uniform vertical electric field follows:

- A. Circular path
- B. Parabolic path
- C. Straight path
- D. Elliptical path

35.

An electron is accelerated through a potential difference V . Its kinetic energy gained is:

- A. eV
- B. e/V
- C. V/e
- D. eV^2

36.

One electron volt is equal to:

- A. $1.6 \times 10^{-19}J$
- B. $1.6 \times 10^{19}J$
- C. $9.1 \times 10^{-31}J$
- D. $1J$

37.

An electron accelerated through $100V$ gains energy:

- A. $100J$
- B. $100eV$
- C. $10eV$
- D. $1eV$

38.

An electron accelerated through $1V$ gains kinetic energy:

- A. $1eV$
- B. $1J$

- C. $1.6 \times 10^{-19} eV$
- D. $9.1 \times 10^{-31} J$

39.

The speed acquired by an electron accelerated from rest through V volts is proportional to:

- A. V
- B. V^2
- C. $\sqrt{V}$
- D. $1/V$

40.

If the accelerating voltage of an electron is increased four times, its speed becomes:

- A. 2 times
- B. 4 times
- C. 8 times
- D. Half

Section E: Electron in Magnetic Field

41.

The magnetic force on a moving electron is:

- A. eE
- B. $evB \sin \theta$
- C. eB/v
- D. ev/E

42.

The magnetic force on a charged particle does:

- A. Positive work
- B. Negative work
- C. Zero work
- D. Infinite work

43.

A charged particle moving perpendicular to a uniform magnetic field follows:

- A. Straight path
- B. Circular path

- C. Parabolic path
- D. Elliptical path

44.

The radius of the circular path of an electron is:

- A. $r = \frac{eB}{mv}$
- B. $r = \frac{mv}{eB}$
- C. $r = \frac{mB}{ev}$
- D. $r = \frac{ev}{mB}$

45.

The angular velocity of an electron in a uniform magnetic field is:

- A. eB/m
- B. m/eB
- C. $e/(mB)$
- D. mB/e

46.

The time period of circular motion of an electron is:

- A. $2\pi m/eB$
- B. $2\pi e/mB$
- C. $eB/2\pi m$
- D. $mB/2\pi e$

47.

The frequency of an electron in a magnetic field is:

- A. $eB/2\pi m$
- B. $2\pi m/eB$
- C. m/eB
- D. e/mB

48.

The radius of an electron's circular path is directly proportional to:

- A. B
- B. $1/B$

- C. B^2
- D. $1/v$ only

49.

If the velocity of an electron is doubled, its circular orbit radius:

- A. Becomes half
- B. Doubles
- C. Becomes four times
- D. Remains unchanged

50.

If the magnetic field is doubled, the radius of the electron's orbit:

- A. Doubles
- B. Becomes half
- C. Becomes four times
- D. Remains unchanged

Section F: Advanced Magnetic-Field Concepts

51.

An electron entering a magnetic field parallel to the field moves in:

- A. Circular path
- B. Helical path
- C. Straight line
- D. Parabolic path

52.

An electron entering a magnetic field at 90° to the field moves in:

- A. Straight line
- B. Circular path
- C. Helical path
- D. Parabolic path

53.

An electron entering a magnetic field at an angle other than 0° or 90° follows:

- A. Circular path
- B. Straight path

- C. Helical path
- D. Parabolic path

54.

In helical motion of an electron in a magnetic field, the component of velocity parallel to B :

- A. Changes continuously
- B. Remains constant
- C. Becomes zero
- D. Reverses direction

55.

The pitch of the helical path is:

- A. $v_{\perp}T$
- B. $v_{\parallel}T$
- C. $vT/2$
- D. $v_{\parallel}/T$

56.

The magnetic field changes the electron's:

- A. Speed only
- B. Kinetic energy only
- C. Direction of velocity
- D. Mass

57.

An electron and proton enter the same magnetic field with equal speeds perpendicular to B . The ratio of their radii is approximately:

- A. 1:1
- B. 1:1837
- C. 1837:1
- D. 1:2

58.

An electron and proton have equal kinetic energy and enter the same magnetic field perpendicular to it. Their radii are proportional to:

- A. m
- B. $1/m$

- C. $\sqrt{m}$
- D. $1/\sqrt{m}$

59.

For equal kinetic energies in the same magnetic field, the proton's radius compared with the electron's radius is approximately:

- A. 43 times
- B. 1836 times
- C. 1/43 times
- D. Equal

60.

The cyclotron frequency of a charged particle is independent of:

- A. Charge
- B. Mass
- C. Magnetic field
- D. Speed

Section G: Millikan's Oil-Drop Experiment

61.

Millikan's experiment was used to determine:

- A. e/m
- B. e
- C. m only
- D. G

62.

The oil used in Millikan's experiment should have:

- A. High evaporation rate
- B. Low evaporation rate
- C. High electrical conductivity
- D. Zero density

63.

The atomizer is used to:

- A. Produce oil droplets
- B. Produce X-rays

- C. Measure charge
- D. Measure voltage

64.

X-rays in Millikan's experiment are used to:

- A. Heat the oil
- B. Ionize the air
- C. Produce light
- D. Increase oil density

65.

The charge on an oil drop is given by:

- A. $q = ne$
- B. $q = n/e$
- C. $q = e/n$
- D. $q = n + e$

66.

The quantization of charge means:

- A. Charge is continuously variable
- B. Charge exists only in integral multiples of e
- C. Charge cannot be measured
- D. Charge is always positive

67.

An oil drop has charge $4.8 \times 10^{-19}C$. The number of excess electrons is:

- A. 2
- B. 3
- C. 4
- D. 5

68.

An oil drop carries $8.0 \times 10^{-19}C$ charge. The number of excess electrons is:

- A. 2
- B. 4
- C. 5
- D. 8

69.

If an oil drop is stationary between two horizontal plates, then:

- A. $qE = mg$
- B. $qE > mg$
- C. $qE < mg$
- D. $qE = 0$

70.

If an oil drop has negative charge and is suspended against gravity, the electric field is directed:

- A. Upward
- B. Downward
- C. Horizontally
- D. Zero

Section H: Millikan Numerical and Conceptual MCQs

71.

An oil drop of mass $2 \times 10^{-14} \text{ kg}$ is suspended in an electric field of $2.5 \times 10^5 \text{ N/C}$. Its charge is approximately:

- A. $7.84 \times 10^{-19} \text{ C}$
- B. $7.84 \times 10^{-20} \text{ C}$
- C. $8.0 \times 10^{-19} \text{ C}$
- D. $3.2 \times 10^{-19} \text{ C}$

72.

The charge calculated in Question 71 corresponds approximately to:

- A. 2 electrons
- B. 3 electrons
- C. 5 electrons
- D. 10 electrons

73.

If the electric field required to suspend an oil drop is doubled while its charge remains constant, the weight of the drop must be:

- A. Doubled
- B. Halved

- C. Unchanged
- D. Four times

74.

If the charge on a suspended oil drop is doubled without changing its mass, the electric field required for suspension becomes:

- A. Double
- B. Half
- C. Four times
- D. Unchanged

75.

If an oil drop is falling with terminal velocity and the electric field is applied upward, the electric force on a negatively charged drop is:

- A. Upward
- B. Downward
- C. Horizontal
- D. Zero

76.

For a negatively charged oil drop to be suspended against gravity, the electric field must be:

- A. Upward
- B. Downward
- C. Horizontal
- D. Zero

77.

The elementary charge e determined by Millikan is approximately:

- A. $1.6 \times 10^{-19} C$
- B. $9.1 \times 10^{-31} C$
- C. $1.76 \times 10^{11} C/kg$
- D. $6.63 \times 10^{-34} C$

78.

If a drop has charge $3.2 \times 10^{-18} C$, the number of excess electrons is:

- A. 2
- B. 10

C. 20

D. 32

79.

The main conclusion of Millikan's experiment is that electric charge is:

A. Continuous

B. Quantized

C. Always positive

D. Always negative

80.

If an oil drop has no charge, it cannot be suspended by:

A. Gravity

B. An electric field

C. Air resistance

D. Its own weight

Section I: Competitive Numerical MCQs

81.

An electron is accelerated through 100V. Its kinetic energy is:

A. 100J

B. 100eV

C. 10eV

D. 1eV

82.

An electron accelerated through 100V has kinetic energy in joules equal to:

A. $1.6 \times 10^{-17} J$

B. $1.6 \times 10^{-19} J$

C. $1.6 \times 10^{-21} J$

D. $9.1 \times 10^{-31} J$

83.

The velocity acquired by an electron accelerated from rest through 100V is approximately:

A. $5.93 \times 10^6 m/s$

B. $5.93 \times 10^5 m/s$

- C. $5.93 \times 10^7 m/s$
- D. $5.93 \times 10^4 m/s$

84.

An electron moves with velocity $2 \times 10^7 m/s$ perpendicular to $B = 0.01T$. The radius of its path is approximately:

- A. $1.14 \times 10^{-2} m$
- B. $1.14 \times 10^{-3} m$
- C. $1.14 \times 10^{-4} m$
- D. $1.14 \times 10^{-1} m$

85.

An electron moves perpendicular to a magnetic field of $0.1T$ with speed $10^7 m/s$. The radius of its orbit is approximately:

- A. $5.7 \times 10^{-4} m$
- B. $5.7 \times 10^{-3} m$
- C. $5.7 \times 10^{-2} m$
- D. $5.7 \times 10^{-5} m$

86.

The time period of an electron in a magnetic field of $0.01T$ is approximately:

- A. $3.57 \times 10^{-9} s$
- B. $3.57 \times 10^{-8} s$
- C. $3.57 \times 10^{-7} s$
- D. $3.57 \times 10^{-6} s$

87.

The frequency of revolution of an electron in a magnetic field of $0.1T$ is approximately:

- A. $2.8 \times 10^8 Hz$
- B. $2.8 \times 10^9 Hz$
- C. $2.8 \times 10^7 Hz$
- D. $2.8 \times 10^6 Hz$

88.

An electron enters a magnetic field of $0.02T$ perpendicular to it with speed $4 \times 10^6 m/s$. The radius of its orbit is approximately:

- A. $1.14 \times 10^{-3}m$
- B. $1.14 \times 10^{-2}m$
- C. $1.14 \times 10^{-4}m$
- D. $1.14 \times 10^{-1}m$

89.

An electron accelerated through $400V$ has a speed compared with one accelerated through $100V$ that is:

- A. 2 times
- B. 4 times
- C. 8 times
- D. Same

90.

An electron moving in a circular path in a magnetic field has kinetic energy K . If its kinetic energy becomes $4K$, its radius becomes:

- A. $r/4$
- B. $r/2$
- C. $2r$
- D. $4r$

Section J: High-Level CEE/IOE Challenge Questions

91.

An electron and a proton are accelerated through the same potential difference. The ratio of their speeds v_e/v_p is approximately:

- A. 1
- B. 43
- C. 1836
- D. $1/43$

92.

An electron and proton accelerated through the same potential difference enter the same magnetic field perpendicular to it. The ratio of their circular radii r_e/r_p is approximately:

- A. 1
- B. 43
- C. $1/43$
- D. 1837

93.

An electron and proton have equal momentum and enter the same magnetic field perpendicular to it. Their circular radii are:

- A. Equal
- B. In ratio 1837: 1
- C. In ratio 1: 1837
- D. In ratio 43: 1

94.

An electron enters crossed electric and magnetic fields undeflected. If E is doubled and B remains unchanged, to remain undeflected its speed must:

- A. Double
- B. Become half
- C. Remain unchanged
- D. Become four times

95.

An electron moves in a circular path in a uniform magnetic field. If its speed is doubled, its angular velocity:

- A. Doubles
- B. Becomes half
- C. Remains unchanged
- D. Becomes four times

96.

An electron moving perpendicular to a magnetic field has radius r . If both its velocity and magnetic field are doubled, the new radius is:

- A. r
- B. $2r$
- C. $r/2$
- D. $4r$

97.

An electron enters a uniform magnetic field with velocity having components $v_{\parallel}$ and $v_{\perp}$. Which component determines the radius of the helix?

- A. $v_{\parallel}$
- B. $v_{\perp}$